Image Transmission Performance Over UHF Channel Using Real Time Processing in LOS and NLOS Transmission

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***Abstract*—Data transmission from the transmitter to the receiver is influenced by several parameters including distance, obstacle and data size. In general, the farther the distance of the transmitter to the receiver, the longer the data will be sent. This relationship is also applied to data size parameter, where the larger the data sent, the longer the data received at the receiver. Therefore, this research aims to measure transmission performance of big size data such as image in real time processing unit applied in Line-of-Sight (LOS) and Non-Line-of-Sight (NLOS) UHF radio module. Since image data processing takes a significant time, the design considered a high-speed processor. Payload, designed using Raspberry Pi Zero W processors and Raspberry Pi Camera Rev1.3, has a function of capturing image data remotely. This payload will communicate with a receiver station over 3DR as Radio Transceiver Module. Receiver station set up to display the received image data. The proposed design performed that the process time of image data transmission is getting longer along with changes in data size and distance. The research results that NLOS data transmission takes longer time compared to LOS data transmission at around 18.483 seconds for 512x384 pixel image with quantization level of K=32. The test conducted to the same image showed that the data loss of NLOS transmission is 8% larger than LOS data transmission.**

Keywords—Line of Sight, Non-Line of Sight, Image Transmission, PSNR, UHF Transmission

I. Introduction

Remote data transmission has been applied for various purposes such as monitoring or controlling functions. Some research has been implementing distance data transmission in some areas including telemetry for measuring environmental parameters [1], energy transfer for smart grid technologies implementation [2], transportation [3] and other areas. Long distance causes a weak signal strength captured by the receiver antenna, which this potentially leads to data loss during transmission process. Data size sent has been affected transmission duration and processing delay that occurred either on the payload or on the receiver segment, as well as environmental terrains such as trees, walls, and other barriers [3]. Besides the obstacle, the characteristics of frequency band of the device also affects system performances [4].

LOS communication that is usually applied to a vertical communication technology is an optimal condition where there is no obstacles between the two connected sub-systems. A research has been implementing this type of communication to the development of space communication [5]. Whereas NLOS communication has been more applied to a horizontal communication where there is a barrier on the communication lines. Communication technology implementing NLOS including RAN technology, UHF transmission such as TV broadcasting, Wi-Fi, and so on. A study has been exploiting the combination of LOS and NLOS for analyzing the proposed system performance [6].

This research focus on designing a device and measure its performance in both transmission environment for big size data (image) considering several parameters including resolution, transmission duration and data loss.

II. Methodology

A. Design Approach

Generally, the proposed transmission device consists of a receiver station and payload. The receiver station serves as a data receiver and user interface. Meanwhile the payload is a transmitter device that functions to capture image and to transmit it to the receiver.

Receiver station consists of processing device connected to UHF transceiver module to be able to communicate with the payload as shown in Figure 1. The receiver station also has a graphics user interface (GUI) to display image data monitoring systems.

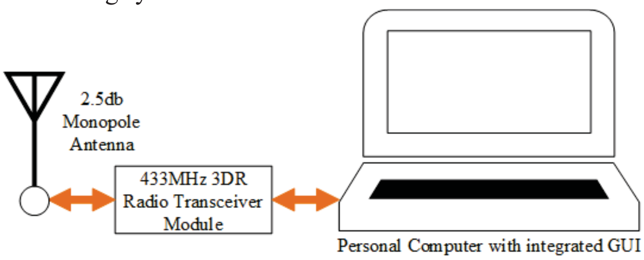

Fig 1. Receiver station block diagram consists of radio transceiver module and PC as user interface

The designed payload, seen in Fig 2, is a transmitter subsystem consists of a processor with a camera. During the process of image recording, the captured image generated by the Pi rev 1.3 camera Raspberry is sent to minicomputer Raspberry Pi 3 and is transferred to the receiver station to be displayed on GUI.

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Back-end software is a program that runs on a microprocessor contained in the Payload. Payload enables remote control process either preconfigured action (already defined automatically and instantly run in the first booting of the microprocessor) or reconfigured action by command received (need reboot) from the receiver station.

GUI as front-end Programming is a software which is directly related to the five senses of human beings that serve as the interface to display any information needed by end users. The retrieval process at the receiver station is represented in Figure 3.

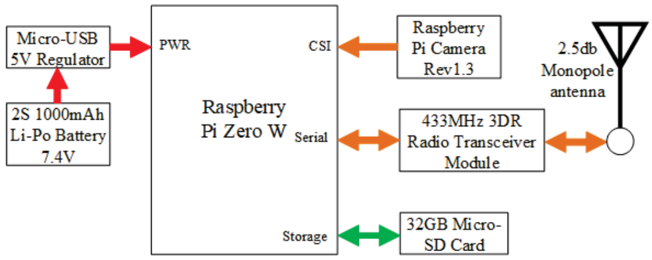

Fig 2. Payload architecture using Raspberry Pi Zero W as proocessor, Raspberry Pi Camera and 3DR as radio tranceiver module

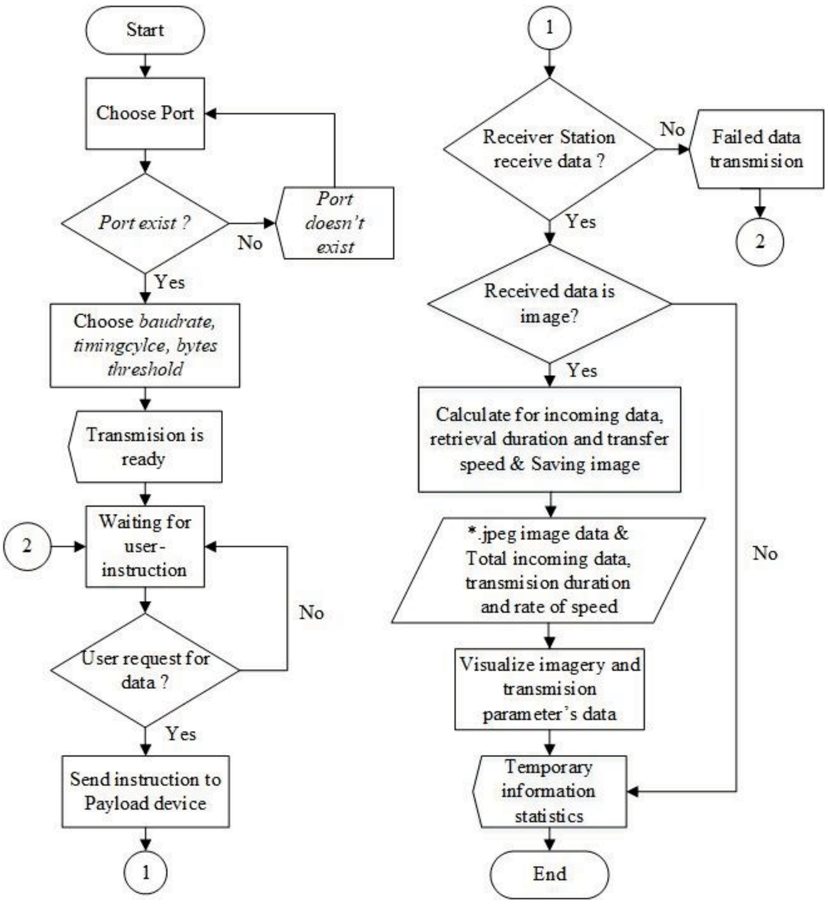

Fig 3. Front-end programming flowchart of receiver station device

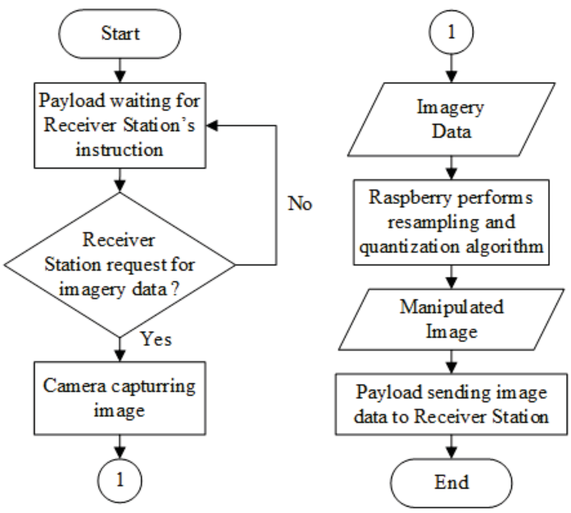

Fig 4. Payload workflow programming

B. Data Conversion

There is a reversible data conversion system needed for image data transmission to receiver station without affecting its data content. The image data should be converted into ASCII format before sending it over UHF wave. This will transform the image data from spatial domain to 8-bit NVT ASCII for each byte of character. The data that are received by receiver will be converted back into spatial domain as a complete image.

C. Digitalizing Image Data

In order to resize the number of image data and its size, this research applies an image processing techniques namely resampling and quantizing process.

C.1. Bilinear Interpolation Image Resampling

Resampling process is a method for remapping image data to new position of coordinate differed from its original image data. This method aims to change image resolution considering the nearest original image aspect ratio compensation. This means that the new output image will have different resolution from the original one.

This research uses bilinear interpolation algorithm to resample image data. Bilinear Interpolation is obtained by averaging 4 nearest pixel using linear interpolation formula to get a new pixel value as represented in Fig 5. By computing this value, bilinear interpolation can be used as a method for either upscaling (increasing image resolution) or downscaling (decreasing image resolution) [7][8].

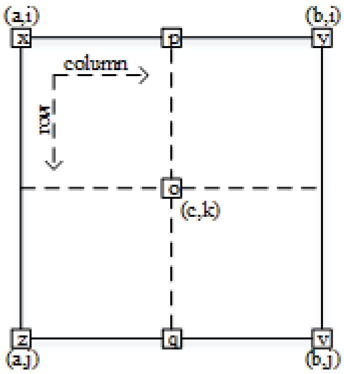

Fig 5. Biliner Interpolation Scheme

As represented in Fig 5, x, y, z, v are the pixel values of image data and (a,i), (b,i), (a,j), (b,j) are position coordinate of each those values respectively. To obtain the o, interpolation value, at point (c,k), the value of p and q need to be calculated. The formula to find the new value, p and q, are defined in (1) and (2).

$$p = \left(\frac{b-c}{b-a} \times x\right) + \left(\frac{c-a}{b-a} \times y\right) \tag{1}$$

$$q = \left(\frac{b-c}{b-a} \times z\right) + \left(\frac{c-a}{b-a} \times v\right) \tag{2}$$

The interpolation pixel value o can be calculated using (3) formula.

$$o = \left(\frac{j-k}{j-i} \times p\right) + \left(\frac{k-i}{j-i} \times q\right) \tag{3}$$

The consequences of manipulating the size of data resolution will result in a change of file size and it is directly proportional to the resampling type. Upscaling resampling will result a file size larger than the original image size, meanwhile the downscaling resampling will have a smaller file size compared to the original one.

C.2. K-means Clusterring Image Color Quantization

The image that has been resampled will be quantized to reduce the color space of the constituent components of the image data. The use of K-means Clustering aims to compress image file size into a smaller one. Reduction of image data on constituent components could be either clearly or vaguely recognized by human interpretation [9].

Generally, the value of K in this method refers to the number of color components which are determined as color combination. The entire color components will be grouped into K different group as expressed in (4).

$$J = \sum_{j=1}^{K} \sum_{i=1}^{n} \left\| c_i - c_j \right\|^2 \tag{4}$$

Where J is an objective function as a result of classification process of the K-means algorithm, K is the number of cluster centers, n is the total number of data components that will be grouped, and $\left\| c_i - c_j \right\|^2$ is a Euclidean distance between c_i and c_j , c_i is the set of data points and c_j is the set of centers.

Fig 6 represented an example of quantization result of sample image using K-means clustering algorithm with K = 2, K = 4 and K = 8.

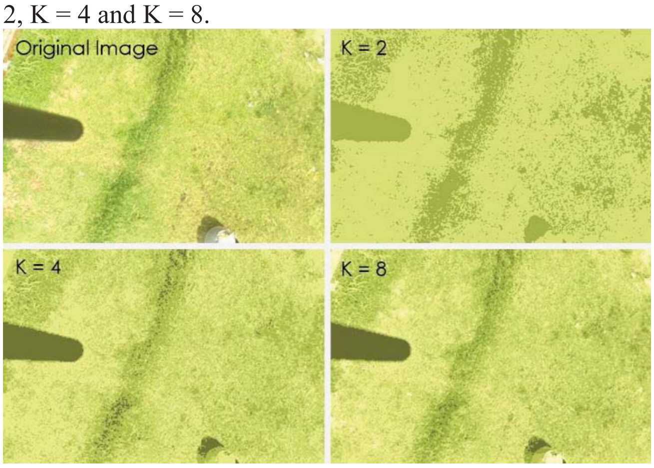

Fig 6. Comparison between a reference image and quantisized images of K=2, K=4 and K=8

D. Transmission Performance

There are several parameters that can be measured in the determination of transmission performance. One of the commonly used is Bit Error Ratio (BER). BER is ratio of the number of bit error compared to the total number of bits as formulated in (5). The number of bit error obtained from total bit which is missing on the process of transmission and total bits that encounter translation error in the receiver station.

$$BER = \frac{n\ Error}{n\ Bits} \tag{5}$$

where $n\ Error$ is the total number of error bits and $n\ Bits$ is the total bit transmitted by the system.

E. Image Quality Assessment

There are several methods to assess image quality namely the MOS, Peak Signal to Noise Ratio (PSNR), Mean Square Error (MSE), etc. This research used PSNR and MSE measure to evaluate image quality.

Mean Square Error (MSE) is the average pixel difference between the processed and original or reference image. MSE value is defined in (6). In this research, the processed image is an image that has been captured by the payload, whereas the reference image is an image that has been received by receiver.

$$MSE = \frac{1}{xy}\sum_{j=1}^{X}\sum_{i=1}^{y}[A(i,j) - B(i,j)]^2 \quad (6)$$

Where $A(i,j)$ is matrix element of an image captured by the payload, $B(i,j)$ is matrix element of an image received by the GS, and x, y are the dimensions of the images, x is row index of matrix element, y is column index of matrix elements.

In image processing, PSNR is a common technique to measure the quality of reconstructed image beside MSE technique. PSNR value, is generally measured in decibel (dB), is defined in (7) or (8), where 255 is the maximum value of gray scale 8 bit [10] [11].

$$PSNR = 10log_{10}\left[\frac{255^2}{\frac{1}{xy}\sum_{j=1}^{X}\sum_{i=1}^{y}[A(i,j)-B(i,j)]^2}\right] \quad (7)$$

$$PSNR = 20\log_{10}\left(\frac{255}{\sqrt{MSE}}\right) \quad (8)$$

From the relation of MSE, PSNR and performance, it is defined that the lower the MSE is (the higher PSNR), the better performance would be.

III. Result and Discussion

Image data retrieval has conducted to test system performances using a combination of resampling and quantization process. The test images are using the sampling resolution of 64x48, 128x96, 256x192, 512x384 pixels, and also using quantization variable of k = 16, k = 32, k =64. The test has been carried out in the variation of the distance ranging from 5-30 meters with two environments, LOS and NLOS, shown in Fig 7. The data has been analyzed using some calculation of BER, PSNR, MSE, transmission duration.

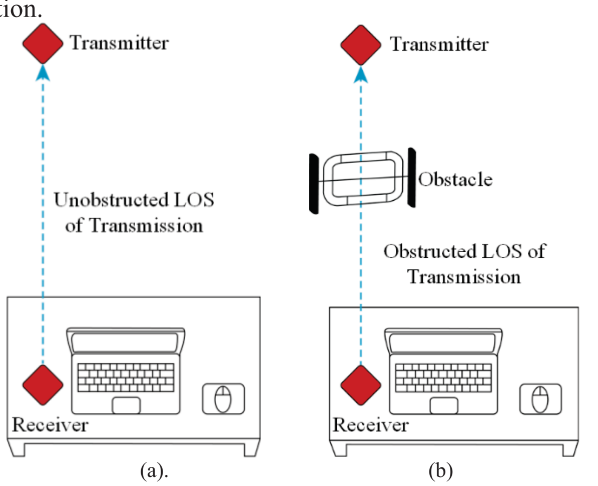

Fig 7. Testing Environment of (a). LOS and (b) NLOS

The LOS data testing is conducted by placing the payload and receiver station in the condition where no obstruction between them. This condition can be assumed as an ideal simulation to reduce signal loss during propagating in the air. The same testing has been carried out to NLOS environment by placing an extra barrier in the form of iron trellis with a goal to disrupts communication lines between the two subsystems.

The image data that will be processed are represented in the form of ASCII data types. The larger image resolution, the greater the number of ASCII characters, as shown in table 1.

TABLE I. The Number of Characters for Each Resolution Sizes

No.	Resolution (Pixel)	Number of characters
1	64x48	2254
2	128x96	7168
3	256x192	26605
4	512x384	86673

A. Comparison between Distance Against The Transmission Duration

The first test is carried out to measure the relationship between the transmission duration on the four types of image data with different resolutions. The graph of Fig 8 has shown that transmission duration of image resolution of 64x48, 128x96 and 256x192 take less than 5 seconds, but different result obtained from image resolution of 512x384, which transmission duration larger than those other image resolutions with ranging of 12-18 seconds. It can be concluded from the graph that the greater the data size (image resolution), the longer the transmission duration.

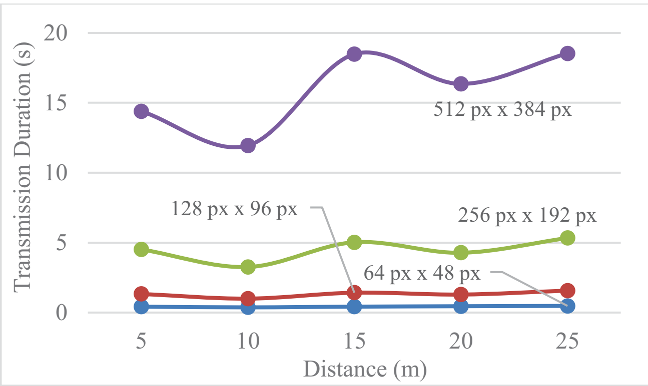

Fig 8. Performance comparison in the parameter of distance againts transmission duration for different size of image resolution

Taking 5 samples of image data 512x384 to be transmitted, the result has been represented in Fig 9 which showed that:

1. The farther the distance between two transceivers, the longer the duration transmission either for LOS or NLOS environment.
2. Transmission duration for NLOS transmission is more fluctuated compared to LOS transmission, since there is more data loss occurred for that NLOS path.

The research result represented that the transmission duration is not only influenced by the distance between transceivers and data size, but also effected by the transmission environment (LOS or NLOS).

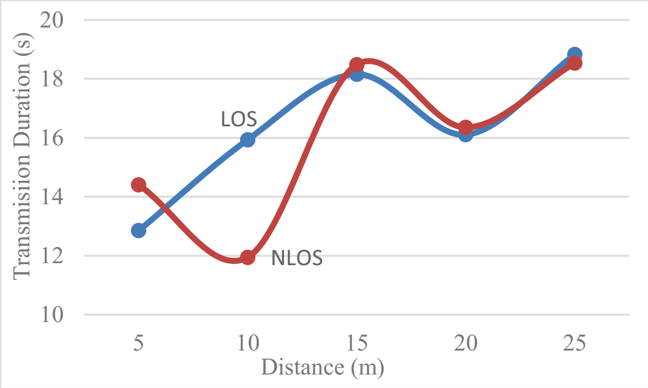

Fig 9. Performance comparison between distance over transmission duration in LOS and NLOS environment

B. *Comparison of transmission LOS and NLOS towards distance and BER*

The second testing is performed to measure LOS and NLOS transmission performance against data loss (BER value). Fig 10 depicted that data loss has significantly occured in distance more than 15 metres. The graph has also shown that BER value of NLOS transmission is always higher than that of LOS path.

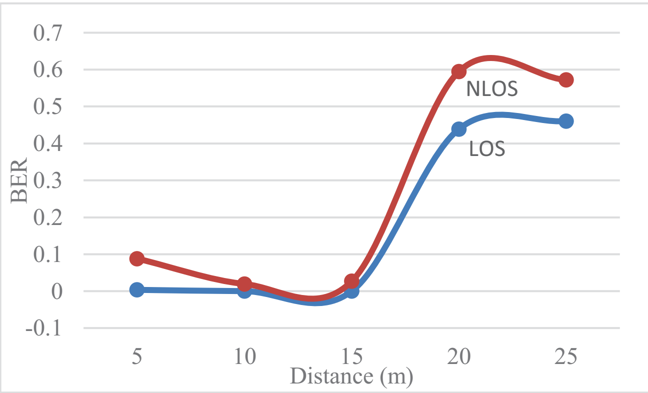

Fig 10. Performance comparison between LOS and NLOS data transmission over distance and BER value

C. *Bit Error Ratio of Image Data Transmission*

Bit Error Ratio is evaluated by comparing the number of data (in bits units) transmitted (the amount of image data captured by the payload) with the number of bits received (the amount of data received on the receiver station). The calculation results are shown in the table 2.

TABLE II. THE BER PERCENTAGE OF IMAGE DATA OF LEVEL QUANTIZATION K=32 IN DISTANCE OF 15M (NLOS PATH)

No.	Total bit sent by the payload	Total bit received by the receiver station	BER
1	21952	21928	5.32%
2	69576	69552	0.69%
3	234896	234856	0.23%
4	855880	855856	0.06%

D. *PSNR and MSE value againts transmission distance*

Image quality can be evaluated using PSNR and MSE value by comparing image data saved in the payload with image revieved by the receiver station. The transmission over LOS path run optimal in the distance up to 15m since there is no error during transmission. This is proven by the MSE of 0, shown in Tabel III, which means that the image transmitted by the payload is identical with the image received in the receiver station. Meanwhile in NLOS path, there will be always data loss during transmission.

TABLE III. THE MSE AND PSNR VALUE OF LOS AND NLOS DATA TRANSMISSION IN THE DIFFERENT DISTANCE OF PAYLOAD AND RECEIVER STATION

Distance (m)	LOS		NLOS	
	MSE	PSNR (dB)	MSE	PSNR (dB)
5	0	Inf	1965820079	-44.8046
10	0	Inf	857547458	-41.2017
15	0	Inf	2464122258	-45.7858
20	1760383000	-44.3252	3344027671	-47.1118
25	1230164015	-42.7688	2575235669	-45.9773

IV. CONCLUSION

Several conclusions have been drawn for this research including:

1. NLOS data transmission is not always smaller than LOS data transmission, because data loss in NLOS path is also affected by the obstacle materials used
2. Data loss over NLOS transmission reaches up to 8% greater than that on the LOS path, which measurements are carried out for the same image (quantization level and image resolution)
3. The process of resampling & data quantizing requires a certain amount of time, which is directly proportional to the K value and image resolution.

ACKNOWLEDGMENT

This research has been funded by The Ministry of Research, Technology and Higher Education and has been also supported by Politeknik Negeri Batam. We are really grateful for their financial and facility support.

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